

EMC2T Electromagnet System

Electromagnet with 4-Quadrant Bipolar Power Supply (Optical Access Compatible)

**Precision Magnetic Field Control.
Stable Bipolar Power Delivery.
Built for Research Applications.**

The EMC2T Electromagnet System is a high-performance research-grade electromagnet capable of generating magnetic fields up to 2 Tesla. It is integrated with a 4-quadrant bipolar power supply, enabling smooth polarity reversal and precise field control without switching transients. Designed for laboratory and industrial environments, the system features adjustable pole gaps, optical access compatibility, and efficient water-cooled coils, making it suitable for advanced experimental setups and material characterization.

KEY FEATURES

- **Maximum Magnetic Field**
Generates magnetic fields up to 2 Tesla
- **4-Quadrant Bipolar Power Supply**
Supports both source and sink operation for flexible bipolar control.
- **Smooth Zero-Crossing Polarity Reversal**
Enables seamless polarity switching with stable transition performance.
- **Adjustable Pole Gap**
Provides adjustable pole spacing from 5–50 mm for versatile experimental configurations.
- **High Stability Performance**
Ensures low ripple and noise for stable magnetic field operation.



Notes

- Outputs are current-limited and optimized for precision, not high-power delivery
- Designed specifically for low-noise experimental environments
- Custom channel counts and configurations available

APPLICATIONS

- Quantum device gate control (qubits, superconducting circuits)
- Nanoelectronic and 2D material experiments
- Quantum transport measurements
- Multi-terminal device biasing
- Low-noise precision voltage sourcing
- Automated measurement systems

EMC2T Electromagnet System

SPECIFICATIONS

MAGNET SPECIFICATIONS

Maximum Magnetic Field	2.0 Tesla (at 20 mm pole gap)
Pole Face Diameter	40 mm
Pole Gap	5 – 50 mm (adjustable)
Coil Gap	140 mm
Ampere-Turns	48,000 (Wide band Gap SiC Based power supply)
Maximum Current	70 A
Coil Resistance	0.45 Ω (each coil @ 20°C)
Power Rating	(70 A, 70 A) 5 μ W
Cooling	Water cooled (8–10 L/min @ ~18°C)
Conductor Material	Copper
Core / Yoke Material	Low carbon steel

MECHANICAL SPECIFICATIONS

Dimensions (L x W x H)	940 x 635 x 770 mm
Weight (Magnet)	~530 kg
Weight (Power Supply)	~35 kg
Optical Access	Provided through pole caps

POWER SUPPLY SPECIFICATIONS (EM40)

Type	Bipolar, linear, 4-quadrant DC
Input Supply	3-phase, 4-wire, 415 VAC
Voltage Range	340 – 530 VAC
Frequency	47 – 63 Hz
Operating Modes	Constant Voltage (CV), Constant Current (CC)
Efficiency	Up to 91%
Power Factor	~0.95
Stability	$\leq 0.01\%$
Line Regulation	$< \pm 0.5\%$
Load Regulation	$< \pm 0.5\%$
Cooling	Forced air (fan with filter)

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